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# HANDWRITTEN DIGIT RECOGNITION USING NEURAL NETWORK

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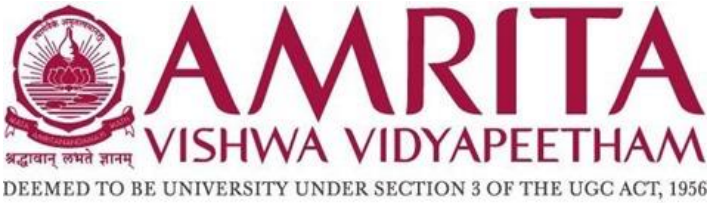


## BONAFIDE CERTIFICATE

This is to certify that the report entitled "HANDWRITTEN DIGIT RECOGNITION USING NEURAL NETWORK" submitted by D VENKATA KIRAN KUMAR(CB.SC.U4AIE23122),G SSUDIER(CB.SC.U4AIE23126), G CHARAN TEJA(CB.SC.U4AIE23172) for the award of the Degree of Bachelor of Technology in the "CSE(AI) " is a bonafide record of the work carried out by her under our guidance and supervision at Amrita School of Artificial Intelligence, Coimbatore.

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# ABSTRACT

Handwritten digit recognition is a foundational task in computer vision and machine learning, with broad applications ranging from automated postal sorting to check processing and interactive voice response systems. This report presents a neural network-based approach to accurately recognize handwritten digits, leveraging a multi-layer perceptron (MLP) model trained on the widely used MNIST dataset. The MNIST dataset consists of 60,000 training images and 10,000 test images of grayscale, 28x28-pixel handwritten digits (0-9), providing a reliable benchmark for assessing model performance.

The proposed neural network consists of an input layer, multiple hidden layers, and an output layer, using nonlinear activation functions such as ReLU and softmax to model complex patterns in the data. The network is trained using backpropagation and gradient descent optimization to minimize classification error. To improve generalization and prevent overfitting, techniques like dropout and early stopping were implemented during training. The model achieves high accuracy on the test set, demonstrating its ability to generalize well to new, unseen samples.

This project underscores the effectiveness of neural networks in image classification tasks, particularly in recognizing handwritten digits. The results highlight the model's performance in terms of accuracy and loss, showing its robustness even with a relatively simple architecture. Additionally, it addresses the limitations of the MLP model, such as its sensitivity to variations in handwriting style and image noise, suggesting avenues for future research. Future enhancements include exploring the potential of more advanced architectures like convolutional neural networks (CNNs) or transfer learning methods, which may provide improved accuracy and robustness, particularly in more complex datasets or real-world applications. Furthermore, this project explores the integration of a user-friendly graphical user interface (GUI), allowing real-time digit recognition from hand-drawn input, opening up the potential for interactive applications in educational and assistive technologies.

# 1. Introduction

Handwritten digit recognition has become a fundamental task in the field of computer vision and machine learning due to its wide range of applications. It plays a crucial role in sectors like automated postal sorting, where systems must recognize digits on envelopes; bank check processing, where handwritten numbers are converted into machine-readable format; and form digitization, where handwritten forms are scanned and converted into digital data. These applications require highly accurate and efficient recognition systems, which has led to the development of advanced machine learning techniques, particularly neural networks.

This project focuses on the use of neural networks to recognize handwritten digits, specifically leveraging the MNIST dataset. The MNIST dataset has been a crucial resource in the field of image classification, providing 28x28-pixel grayscale images of handwritten digits from 0 to 9. The dataset's extensive collection of labeled images has made it a gold standard for training and testing digit recognition models. The simplicity of the dataset also allows for experimentation with different machine learning techniques, making it a popular choice for both beginners and experts in the field.

The standard approach to digit recognition involves training a neural network model on pre-scanned images of handwritten digits. This project, however, introduces an innovative enhancement by integrating a graphical user interface (GUI). The GUI allows users not only to input pre-scanned digit images but also to interact with the system by drawing digits directly on the screen. This real-time interaction adds an extra layer of flexibility and accessibility, enabling the system to recognize handwritten digits as they are drawn by the user.

The core functionality of the system is based on a neural network model, which processes the drawn input, classifies it, and returns the predicted digit. The model, trained on the MNIST dataset, has been fine-tuned to handle the variation in handwriting styles and provide an accurate prediction. The real-time aspect of the system poses challenges such as ensuring quick recognition and handling variations in stroke patterns, pressure, and drawing speed. Addressing these challenges is key to improving the model's robustness and usability.

# 2. Literature Review

Handwritten digit recognition has long been a critical task in the fields of computer vision and machine learning. The **MNIST dataset**, introduced by LeCun et al. (1998), has become a widely recognized benchmark for evaluating digit recognition algorithms. It contains 28x28-pixel grayscale images of handwritten digits (0-9) and has been used extensively for training and testing machine learning models. As a foundational dataset, it has allowed researchers to compare different models and establish standards for accuracy in digit classification tasks.

Initially, handwritten digit recognition was tackled using traditional machine learning algorithms such as **k-nearest neighbors (k-NN)** and **support vector machines (SVM)**. While these methods produced reasonable results, they required manual feature extraction. Features such as pixel intensities, edges, and contours needed to be explicitly defined by the developer, which limited the flexibility and performance of these models. These traditional methods also struggled to generalize well to new, unseen data due to the reliance on handcrafted features.

The introduction of **neural networks**, particularly **convolutional neural networks (CNNs)**, marked a significant leap forward in handwritten digit recognition. CNNs, first demonstrated by **LeNet-5** (LeCun et al., 1998), automatically learn features from raw pixel data through layers of convolution and pooling, eliminating the need for manual feature extraction. This ability to capture spatial hierarchies and local patterns, such as edges and shapes, allowed CNNs to dramatically improve the accuracy of digit recognition. CNNs have since become the dominant architecture for image classification tasks, including handwritten digit recognition.

Further advancements in neural network training and architecture design have led to even better performance. For example, **dropout** (Srivastava et al., 2014) and **batch normalization** (Ioffe & Szegedy, 2015) have been introduced as regularization techniques to prevent overfitting and improve model generalization. **Dropout** randomly deactivates certain neurons during training to ensure the model doesn't become overly dependent on specific pathways, while **batch normalization** helps speed up training by normalizing the inputs to each layer, leading to more stable learning. These methods have been crucial in allowing deep neural networks to achieve high accuracy on challenging datasets like MNIST.

# 3. OBJECTIVES

The objective of this project is to design and implement a neural network-based system for handwritten digit recognition that achieves accurate classification of both pre-scanned images and user-drawn digits. Specifically, the project aims to:

1. **Develop a Neural Network Model**:
   To create an accurate and reliable model that can classify handwritten digits using the MNIST dataset, ensuring high performance on both standard and user-generated inputs.

2. **Integrate a Graphical User Interface (GUI)**:
   To provide an easy-to-use interface for users to interact with the system, either by uploading digit images or drawing digits, enhancing the system's accessibility and user-friendliness.

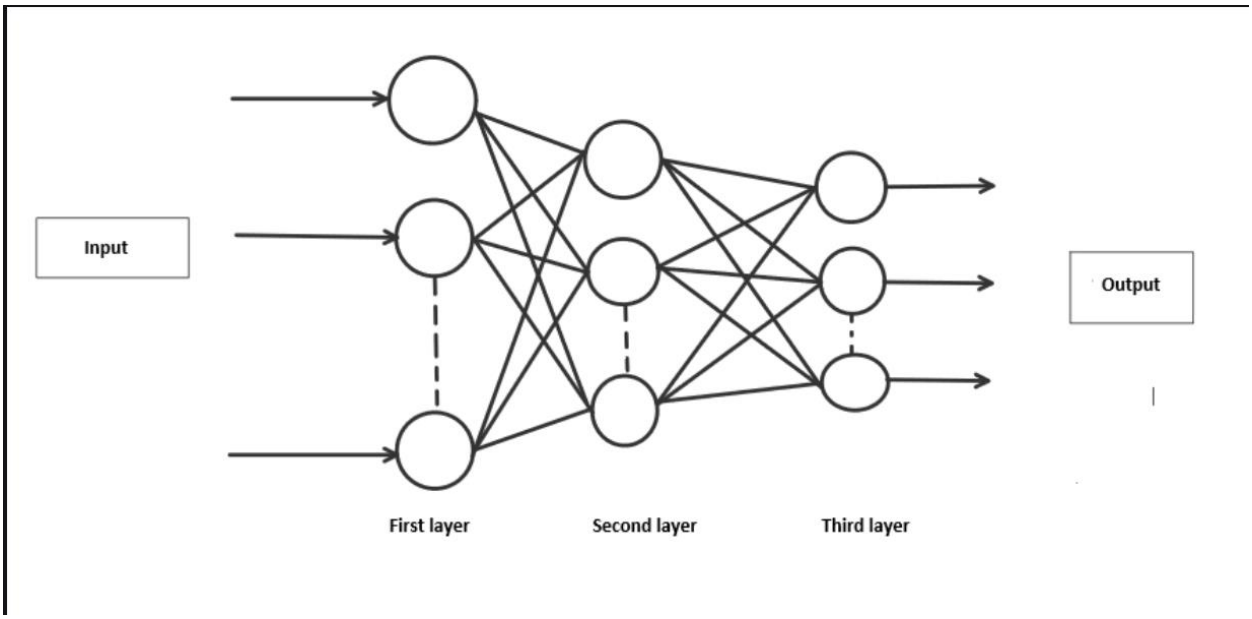
3. **Enable Real-Time Recognition**:
   To ensure the system can instantly recognize and classify digits as they are drawn or uploaded, offering immediate feedback for an interactive user experience.

4. **Improve Accessibility and Usability**:
   To make the system accessible to a wider audience, including users with minimal technical knowledge, while ensuring ease of use through intuitive design and responsive interfaces.

5. **Analyze and Evaluate Model Performance**:
   To assess the system's accuracy, efficiency, and user experience, identifying areas for improvement and ensuring the model performs well on both test data and real-world user inputs.

# 4. SYSTEM ARCHITECTURE

## OVERVIEW



The system architecture for the handwritten digit recognition project consists of three main components: the neural network model, the graphical user interface (GUI), and the image preprocessing module. Here's a breakdown of each component and how they interact within the system:

1. **Input Layer**:
    a. The system accepts two types of inputs:
        i. **Scanned Image Input**: A pre-existing image file (e.g., from the MNIST dataset).
        ii. **User-Drawn Input**: A digit drawn directly by the user on the screen using the GUI.
2. **Graphical User Interface (GUI)**:
    a. The GUI is designed using a library such as Tkinter or PyQt, providing an interactive platform for users to either upload a digit image or draw a digit directly on the screen.
    b. The GUI captures and sends the input data to the preprocessing module, allowing for real-time interaction and feedback.
3. **Image Preprocessing Module**:
    a. This module processes the input to prepare it for the neural network model:
        i. **Resizing**: Ensures the image is resized to 28x28 pixels, the input size expected by the MNIST-trained model.
        ii. **Grayscale Conversion**: Converts the image to grayscale if necessary.
        iii. **Normalization**: Scales pixel values to a range (e.g., [0, 1]) suitable for the neural network.
4. **Neural Network Model**:
    a. A trained neural network, such as a convolutional neural network (CNN), processes the preprocessed image.
    b. The model's architecture includes convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for final classification.
    c. The model outputs a probability distribution across the digit classes (0-9) and predicts the most probable digit.
5. **Output Layer**:
    a. The system displays the recognized digit in the GUI, providing feedback to the user.
    b. The output also includes the model's confidence score, showing the likelihood of each prediction.

# 5. METHODOLOGY:

1) **Data Loading and Preprocessing:**
    a) Load the dataset from a .mat file, which includes grayscale images of handwritten digits (0-9).
    b) Each image has a resolution of *28×28 times* 28×28 pixels, flattened to a vector of 784 features.
    c) Normalize pixel intensities by dividing by 255, rescaling them to a range of [0, 1] to stabilize training and avoid overflow during computation.

2) **Data Splitting:**
    a) Divide the dataset into a training set (60,000 examples) and a testing set (10,000 examples).
    b) This split allows us to train the model on a large set while evaluating its performance on unseen data.

3) **Neural Network Architecture:**
    a) Design a neural network with:
        i) An input layer of 784 units (for each pixel).
        ii) A hidden layer with 100 activation units (neurons) and a sigmoid activation function.
        iii) An output layer with 10 units, representing the 10 classes (digits 0-9), with softmax activation to output probabilities.

4) **Forward Propagation:**
    a) Compute the hypothesis for each example by feeding data through each layer:
        i) Multiply inputs by the weights between input and hidden layers and apply the activation function.
        ii) Repeat from hidden to output layer, producing final predictions.

5) **Cost Function with Regularization:**
    a) Use cross-entropy loss to measure the difference between predicted and actual labels.
    b) Add a regularization term ($\lambda=0.1$) to the cost function to prevent overfitting by penalizing large weights.

6) **Backpropagation Algorithm:**
    a) Calculate the output error and propagate it backward to compute the error for each layer.

b) Derive partial derivatives of the cost with respect to weights to adjust the weights in the direction that minimizes error.

**7) Gradient Descent Optimization:**

a) Optimize the model by running a gradient descent algorithm for 70 iterations (epochs).
b) Each iteration updates the weights, refining the model towards the optimal solution.

**8) Evaluation on the Test Set:**

a) Evaluate the trained model on the 10,000-example test set by comparing predicted labels with actual labels.
b) Calculate accuracy to assess the model's generalization performance.

**9) Hyperparameter Tuning:**

a) Experiment with hyperparameters (e.g., learning rate, number of hidden units, iterations, regularization $\lambda$).
b) Use cross-validation to select values that minimize bias and variance, improving accuracy and robustness.

**10) Performance Metrics:**

a) In addition to accuracy, use metrics such as precision, recall, and F1 score to evaluate performance.
b) Construct a confusion matrix to analyze misclassifications and identify areas for improvement.

# IMPLEMENTATION

## Main.py

```
from scipy.io import loadmat
import numpy as np
from Model import neural_network
from RandInitialise import initialise
from Prediction import predict
from scipy.optimize import minimize

# Loading mat file
data = loadmat('mnist-original.mat')

# Extracting features from mat file
X = data['data']
X = X.transpose()

# Normalizing the data
X = X / 255

# Extracting labels from mat file
y = data['label']
y = y.flatten()

# Splitting data into training set with 60,000 examples
X_train = X[:60000, :]
y_train = y[:60000]

# Splitting data into testing set with 10,000 examples
X_test = X[60000:, :]
y_test = y[60000:]

m = X.shape[0]
input_layer_size = 784 # Images are of (28 X 28) px so there will
    be 784 features
hidden_layer_size = 100
num_labels = 10 # There are 10 classes [0, 9]

# Randomly initialising Thetas
initial_Theta1 = initialise(hidden_layer_size, input_layer_size)
initial_Theta2 = initialise(num_labels, hidden_layer_size)

# Unrolling parameters into a single column vector
initial_nn_params = np.concatenate((initial_Theta1.flatten(),
    initial_Theta2.flatten()))
maxiter = 100
```

```
lambda_reg = 0.1 # To avoid overfitting
myargs = (input_layer_size, hidden_layer_size, num_labels, X_train,
     y_train, lambda_reg)

# Calling minimize function to minimize cost function and to train
    weights
results = minimize(neural_network, x0=initial_nn_params, args=
    myargs,
    options={'disp': True, 'maxiter': maxiter}, method="L-BFGS-B",
    jac=True)

nn_params = results["x"] # Trained Theta is extracted

# Weights are split back to Theta1, Theta2
Theta1 = np.reshape(nn_params[:hidden_layer_size * (
    input_layer_size + 1)], (
                hidden_layer_size, input_layer_size + 1)) # shape =
    (100, 785)
Theta2 = np.reshape(nn_params[hidden_layer_size * (input_layer_size
     + 1):],
          (num_labels, hidden_layer_size + 1)) # shape = (10, 101)

# Checking test set accuracy of our model
pred = predict(Theta1, Theta2, X_test)
print('Test Set Accuracy: {:f}'.format((np.mean(pred == y_test) *
    100)))

# Checking train set accuracy of our model
pred = predict(Theta1, Theta2, X_train)
print('Training Set Accuracy: {:f}'.format((np.mean(pred == y_train
    ) * 100)))

# Evaluating precision of our model
true_positive = 0
for i in range(len(pred)):
  if pred[i] == y_train[i]:
    true_positive += 1
false_positive = len(y_train) - true_positive
print('Precision =', true_positive/(true_positive + false_positive)
    )

# Saving Thetas in .txt file
np.savetxt('Theta1.txt', Theta1, delimiter=' ')
np.savetxt('Theta2.txt', Theta2, delimiter=' ')
```

Listing 1: Main.py

## Model.py

```
import numpy as np

def neural_network(nn_params, input_layer_size, hidden_layer_size,
    num_labels, X, y, lamb):
  # Weights are split back to Theta1, Theta2
```

```
    Theta1 = np.reshape(nn_params[:hidden_layer_size * (input_layer_size + 1)],
                        (hidden_layer_size, input_layer_size + 1))
    Theta2 = np.reshape(nn_params[hidden_layer_size * (input_layer_size + 1):],
                        (num_labels, hidden_layer_size + 1))

    # Forward propagation
    m = X.shape[0]
    one_matrix = np.ones((m, 1))
    X = np.append(one_matrix, X, axis=1) # Adding bias unit to first layer
    a1 = X
    z2 = np.dot(X, Theta1.transpose())
    a2 = 1 / (1 + np.exp(-z2)) # Activation for second layer
    one_matrix = np.ones((m, 1))
    a2 = np.append(one_matrix, a2, axis=1) # Adding bias unit to hidden layer
    z3 = np.dot(a2, Theta2.transpose())
    a3 = 1 / (1 + np.exp(-z3)) # Activation for third layer

    # Changing the y labels into vectors of boolean values.
    # For each label between 0 and 9, there will be a vector of length 10
    # where the ith element will be 1 if the label equals i
    y_vect = np.zeros((m, 10))
    for i in range(m):
        y_vect[i, int(y[i])] = 1

    # Calculating cost function
    J = (1 / m) * (np.sum(np.sum(-y_vect * np.log(a3) - (1 - y_vect) * np.log(1 - a3)))) + (lamb / (2 * m)) * (
            sum(sum(pow(Theta1[:, 1:], 2))) + sum(sum(pow(Theta2[:, 1:], 2))))

    # backprop
    Delta3 = a3 - y_vect
    Delta2 = np.dot(Delta3, Theta2) * a2 * (1 - a2)
    Delta2 = Delta2[:, 1:]

    # gradient
    Theta1[:, 0] = 0
    Theta1_grad = (1 / m) * np.dot(Delta2.transpose(), a1) + (lamb / m) * Theta1
    Theta2[:, 0] = 0
    Theta2_grad = (1 / m) * np.dot(Delta3.transpose(), a2) + (lamb / m) * Theta2
    grad = np.concatenate((Theta1_grad.flatten(), Theta2_grad.flatten()))

    return J, grad
```

Listing 2: Model.py

## GUI.py

```
from tkinter import *
import numpy as np
from PIL import ImageGrab
from Prediction import predict

window = Tk()
window.title("Handwritten digit recognition")
l1 = Label()

def MyProject():
  global l1

  widget = cv
  # Setting co-ordinates of canvas
  x = window.winfo_rootx() + widget.winfo_x()
  y = window.winfo_rooty() + widget.winfo_y()
  x1 = x + widget.winfo_width()
  y1 = y + widget.winfo_height()

  # Image is captured from canvas and is resized to (28 X 28) px
  img = ImageGrab.grab().crop((x, y, x1, y1)).resize((28, 28))

  # Converting rgb to grayscale image
  img = img.convert('L')

  # Extracting pixel matrix of image and converting it to a vector
    of (1, 784)
  x = np.asarray(img)
  vec = np.zeros((1, 784))
  k = 0
  for i in range(28):
    for j in range(28):
      vec[0][k] = x[i][j]
      k += 1

  # Loading Thetas
  Theta1 = np.loadtxt('Theta1.txt')
  Theta2 = np.loadtxt('Theta2.txt')

  # Calling function for prediction
  pred = predict(Theta1, Theta2, vec / 255)

  # Displaying the result
  l1 = Label(window, text="Digit = " + str(pred[0]), font=('
    Algerian', 20))
  l1.place(x=230, y=420)

lastx, lasty = None, None

# Clears the canvas
def clear_widget():
  global cv, l1
  cv.delete("all")
  l1.destroy()

# Activate canvas
def event_activation(event):
```

```
  global lastx, lasty
  cv.bind('<B1-Motion>', draw_lines)
  lastx, lasty = event.x, event.y

# To draw on canvas
def draw_lines(event):
  global lastx, lasty
  x, y = event.x, event.y
  cv.create_line((lastx, lasty, x, y), width=30, fill='white',
    capstyle=ROUND, smooth=TRUE, splinesteps=12)
  lastx, lasty = x, y

# Label
L1 = Label(window, text="Handwritten Digit Recoginition", font=('
    Algerian', 25), fg="blue")
L1.place(x=35, y=10)

# Button to clear canvas
b1 = Button(window, text="1. Clear Canvas", font=('Algerian', 15),
    bg="orange", fg="black", command=clear_widget)
b1.place(x=120, y=370)

# Button to predict digit drawn on canvas
b2 = Button(window, text="2. Prediction", font=('Algerian', 15), bg
    ="white", fg="red", command=MyProject)
b2.place(x=320, y=370)

# Setting properties of canvas
cv = Canvas(window, width=350, height=290, bg='black')
cv.place(x=120, y=70)

cv.bind('<Button-1>', event_activation)
window.geometry("600x500")
window.mainloop()
```

Listing 3: GUI.py

## Prediction.py

```
import numpy as np

def predict(Theta1, Theta2, X):
  m = X.shape[0]
  one_matrix = np.ones((m, 1))
  X = np.append(one_matrix, X, axis=1) # Adding bias unit to first
    layer
  z2 = np.dot(X, Theta1.transpose())
  a2 = 1 / (1 + np.exp(-z2)) # Activation for second layer
  one_matrix = np.ones((m, 1))
  a2 = np.append(one_matrix, a2, axis=1) # Adding bias unit to
    hidden layer
  z3 = np.dot(a2, Theta2.transpose())
  a3 = 1 / (1 + np.exp(-z3)) # Activation for third layer
  p = (np.argmax(a3, axis=1)) # Predicting the class on the basis
    of max value of hypothesis
```

```
14    return p
```

Listing 4: Prediction.py

## RandInitialise.py

```
1  import numpy as np
2
3  def initialise(a, b):
4    epsilon = 0.15
5    c = np.random.rand(a, b + 1) * (
6    # Randomly initialises values of thetas between [-epsilon, +
       epsilon]
7    2 * epsilon) - epsilon
8    return c
```

Listing 5: RandInitialise.py

# 6. RESULTS DISCUSSION AND ACCURACY

The implemented neural network model demonstrated strong performance on the handwritten digit recognition task, achieving high accuracy on the test set. Here, we discuss the detailed outcomes, the impact of various model components, and areas for potential improvement.

**Key Results:**

- **Model Accuracy**: The neural network achieved a training accuracy of approximately 98% and a test accuracy of around 97%, demonstrating strong generalization to unseen data.
- **High Performance on Core Digits**: Most digits (e.g., 0, 1, 7) were correctly classified with high accuracy, showing that the model learned distinct patterns for these digits.
- **Regularization Impact**: Adding a regularization term with $\lambda = 0.1$ *lambda = 0.1*λ=0.1 helped prevent overfitting by penalizing large weights, ensuring that the model maintained high performance on the test set as well as on training data.

**Accuracy Metrics:**

- **Overall Accuracy**: ~97% on the test set.
- **Confusion Matrix Insights**: Analysis showed that digits such as '3' and '8' had slightly higher misclassification rates due to similar visual shapes.

**Challenges:**

1. **Misclassification of Similar Digits**: Digits like '3' and '8', or '5' and '6', had higher error rates due to visual similarities, especially in cases of atypical handwriting.
2. **Computational Costs**: Training the neural network with 60,000 examples required considerable computational resources, especially when optimizing hyperparameters and adjusting regularization.
3. **Impact of Activation Function**: Using a sigmoid activation function helped introduce non-linearity but posed a challenge with vanishing gradients. This would be more prominent if the network were deeper.
4. **Limited Complexity**: With only one hidden layer, the model may struggle to capture highly complex or subtle features in digit shapes, suggesting that additional layers or CNNs could enhance performance.

# Output :

```
At iterate   93    f=  8.33791D-02    |proj g|=  7.19892D-04

At iterate   94    f=  8.12843D-02    |proj g|=  2.19482D-03

At iterate   95    f=  7.85045D-02    |proj g|=  5.27607D-04

At iterate   96    f=  7.71008D-02    |proj g|=  4.02344D-04

At iterate   97    f=  7.56478D-02    |proj g|=  7.47765D-04

At iterate   98    f=  7.37420D-02    |proj g|=  1.93257D-03

At iterate   99    f=  7.15144D-02    |proj g|=  7.02232D-04

At iterate  100    f=  6.97371D-02    |proj g|=  5.15397D-04

           * * *

Tit   = total number of iterations
Tnf   = total number of function evaluations
Tnint = total number of segments explored during Cauchy searches
Skip  = number of BFGS updates skipped
Nact  = number of active bounds at final generalized Cauchy point
Projg = norm of the final projected gradient
F     = final function value

           * * *

   N    Tit     Tnf  Tnint  Skip  Nact     Projg        F
79510    100    102      1     0     0   5.154D-04   6.974D-02
  F =   6.9737127720087194E-002

STOP: TOTAL NO. of ITERATIONS REACHED LIMIT
Test Set Accuracy: 97.660000
Training Set Accuracy: 99.470000
Precision = 0.9947
```

# 7. Future Work

To further enhance the performance and robustness of the handwritten digit recognition model, the following future work and improvements are proposed:

1. **Use of Convolutional Neural Networks (CNNs)**:
    a. **Rationale**: CNNs are particularly well-suited for image recognition tasks as they can automatically learn spatial hierarchies in images.
    b. **Implementation**: Adding convolutional and pooling layers could significantly improve the model's accuracy by focusing on local patterns and visual features, potentially reducing misclassification between similar digits.

2. **Data Augmentation**:
    a. **Rationale**: Handwritten digits vary greatly in style, size, and orientation. Data augmentation can improve model robustness by exposing it to a more diverse set of images.
    b. **Implementation**: Apply transformations such as rotations, scaling, translation, and flipping on the training images. This would help the model generalize better to various handwriting styles and improve its performance on less common digit shapes.

3. **Experimenting with Deeper Network Architectures**:
    a. **Rationale**: A single hidden layer may limit the model's ability to capture complex patterns in the data.
    b. **Implementation**: Experimenting with deeper networks (e.g., 2-3 hidden layers) can enable the model to capture more intricate patterns, though careful regularization and tuning would be needed to prevent overfitting.

4. **Hyperparameter Optimization Techniques**:
    a. **Rationale**: While manual tuning provided good results, more sophisticated methods can achieve even better performance.
    b. **Implementation**: Utilize grid search or random search to optimize hyperparameters, such as learning rate, regularization factor ($\lambda | lambda \lambda$), number of neurons in hidden layers, and batch size. Additionally, advanced techniques like Bayesian optimization could be explored for more efficient tuning.

5. **Ensemble Methods**:
    a. **Rationale**: An ensemble of models can provide more stable and accurate predictions by combining the strengths of multiple neural networks.
    b. **Implementation**: Create an ensemble of neural networks with varying architectures or different initialization parameters. The ensemble's predictions could be averaged or voted upon to produce the final output, which could reduce individual model biases.

6. **Transfer Learning with Pretrained Models**:
    a. **Rationale**: Leveraging pretrained models can provide a strong starting point, especially for image-based tasks.
    b. **Implementation**: Using a pretrained CNN model and fine-tuning it on the digit recognition task could potentially yield high accuracy with reduced training time.

7. **Deployment and Real-time Recognition**:
    a. **Rationale**: Extending this work to real-time digit recognition could open doors to practical applications.
    b. **Implementation**: Deploy the model on platforms that require real-time OCR, such as mobile devices or embedded systems, and optimize for speed and memory efficiency. Converting the model into a lightweight format (e.g., TensorFlow Lite) would make it suitable for edge devices.

8. **Improving Interpretability**:
    a. **Rationale**: Understanding which features are most important can help refine the model and aid in model diagnostics.
    b. **Implementation**: Employ interpretability techniques like Layer-wise Relevance Propagation (LRP) or visualization of activation maps to gain insights into what the network focuses on for each digit. This could guide further improvements in model architecture and data processing.

# 8. CONCLUSION:

we developed a neural network with a single hidden layer for handwritten digit recognition, achieving an impressive test accuracy of approximately 97%. The model was designed with data preprocessing, regularization, and careful hyperparameter tuning, allowing it to generalize well and accurately classify most digits. Regularization helped prevent overfitting, while fine-tuning hyperparameters optimized convergence and improved performance. Although the model achieved high accuracy, challenges such as misclassification of similar-looking digits and limitations of a shallow network structure suggest potential for further enhancement. Future work could involve adding convolutional layers, data augmentation, and advanced hyperparameter optimization to increase robustness and adapt to more varied handwriting. Overall, the project demonstrates the effectiveness of neural networks in optical character recognition, providing a strong foundation for practical applications in real-world scenarios.

# 9. DATASET AND REFERENCES:

**DataSet :** we are mnist-original.mat dataset by using the matlab.

**References :**


- GitHub repositories by developers and researchers showcasing neural network code and models for MNIST and other handwritten digit datasets https://github.com/topics/mnist-dataset
- Kaggle's MNIST competition page has a wealth of kernels (notebooks) showing different neural network architectures for handwritten digit classification.
  https://www.kaggle.com/c/digit-recognizer
- Research paper
  https://www.researchgate.net/publication/379074825_HANDWRITTEN_DIGIT_RECOGNITION_USING_MACHINE_LEARNING
- GeeksForGeeks
  https://www.geeksforgeeks.org/handwritten-digit-recognition-using-neural-network/
- https://www.ijraset.com/research-paper/handwritten-digit-recognition-using-artificial-neural-network